S-21-13

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MILITARY SPECIFICATION

COMPUTER, AERIAL PHOTOGRAPHIC DATA

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

*1.1 Scope. This specification covers a slide-rule type, manual computer, designated LM-221A, to be used for solving photographic problems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electro-Deposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Design. The aerial photo slide rule shall have a standard side and an aerial side. The slide rule shall have maximum dimensions of 12.5 x 2.5 x 0.4 inches. All scales with the exception of the CIF and Image Size shall start from the same position.

3.1.1 Aerial photo side.

*3.1.1.1 Left runner. The left runner shall have a Logarithmic Altitude Scale, which shall be calibrated from 100 to 250,000 feet, reading from bottom to top respectively. This scale in actuality is 3 1/4 equal logarithmic scales added together. Numbers shall be placed at the (1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9) x 10P and at 250,000. For example, there would be a number 150, 1,500, 15,000, 150,000 since for this scale p varies from 2 thru 5. The length of each logarithmic scale shall be 6.25 centimeters (cm). This scale's 100 mark shall be a minimum of 1.0 inches from the bottom of the end of the plate. This scale shall be the reference for all other scales and shall be labeled ALTITUDE (FEET) in bold letters and this label shall be placed between the 1,000 and 5,000 feet mark. Above the 250,000 mark, the following note shall be placed in legible size print: SET LENS F/L OPP ALT & READ SCALE, FT/IN 9" x 18" COV OPP. THEIR INDEX MARKS. OPP IMAGE SIZE (CM OR FT) READ GROUND SIZE. IF IMAGE IS MEASURED IN INCHES, USE F/L & ALT SCALE TO GET GROUND SIZE.

3.1.1.2 Right runner. The right runner shall have a Logarithmic Ground Size Scale, which shall be calibrated from 1 to 10,000 feet and from .3 to 3050 meters, both reading from bottom to top. This scale in actuality, is 4 logarithmic scales, the same size as those which make up the altitude scale (see 3.1.1.1). Graduation numbers shall be positioned as specified in the fore referenced paragraph for both feet and meters on each respective scale. The portion of the feet scale from 100 to 1,000 shall be denoted and labeled, GROUND SPEED (MPH OR KNOTS). The feet-meter graduations shall be distinctively labeled, GROUND SIZE (FEET) and GROUND SIZE (METERS) respectively. The meter graduations shall be laid out parallel to the feet graduations so that a direct conversion is possible from feet to meters. An arrow and the number 3.6 shall appear on the feet scale in the proper location and shall be labeled 60% O.L. ON 9". The following note shall be placed on this runner in a suitable location: TO OBTAIN CYCLING TIME FOR 60% O.L. (ON CAMERAS WHEN 9" OF FILM IS L.O.F) SET IMAGE SPEED (IN/SEC) IN CM SCALE OPP. INDEX AT 3.6, THEN READ CYCLING TIME IN SEC. OPP. CYCLING TIME INDEX.

3.1.1.3 Slider

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3.1.1.3.1 Left side. The left side of the slider shall have a focal length scale calibrated in inches covering a range of 1 to 240. The scale shall run from bottom to top of the computer. The focal length scale shall consist of 2 1/4 logarithmic scales the same size as and numbered as those specified in paragraph 3.1.1.1. The following notations shall also be included on the focal length scale: At 2.25, an arrow and the symbols 70mm COV. At 4.5, an arrow and the symbols 4.5" COV. At 6.93, an arrow and the symbol MPH. At 7.98, an arrow and the word KNOTS. At 9, an arrow and the symbol 9" COV. At 12, the number 12, an arrow and the word SCALE. At 18, an arrow and the symbol 18" COV. Also lines and numbers shall be included at 24, 36, 48, 72, 96, 144, and 240. This scale shall be labeled FOCAL LENGTH (inches). The following note shall appear above this label: TO OBTAIN IMAGE SPEED (IN/SEC) SET EITHER KNOTS OR MPH INDEX OPP. CORRECT SCALE VALUE & OPP. GROUND SPEED READ IMAGE SPEED (IN/SEC) ON CM SCALE.

3.1.1.3.2 Right side. The right side of the slider shall have an Image Size Scale calibrated in cm and feet from 0.03 to 100 cm and 0.001 to 4 feet bottom to top respectively. The foot scale shall consist of 3 2/5 equal logarithmic scales as specified in 3.1.1.1. However, whereas the other scales all start from the same basic line once the slide rule is assembled and aligned, this scale starts in feet at the 1.2 mark on the Focal Length Scale. The cm scale shall be positioned parallel to the feet scale so that conversion is direct from cm to feet. Both the cm and feet scale shall be numbered as specified in 3.1.1.1. At the 1 cm mark, an arrow shall be added and the words CYCLING TIME. The double scale shall be labeled IMAGE SIZE and the proper scales labeled "Ft" and "cm".

3.1.1.3.3 At the bottom of the slider, parallel to all graduations, and on a plane with the 1 graduation of the Focal Length Scale, an arrow shall be placed from the right side, over to, but not thru the numeral 1 on the Focal Length Scale. This line shall be labeled FEET PER INCH. When the slide rule is assembled and aligned, this arrow shall appear as a line which extends from the 1 mark on the Ground Size feet scale, and if projected, would pass through the 1 mark of the Focal Length Scale and 100 mark of the Altitude Scale.

3.1.2 Standard slide rule side.

•3.1.2.1 All scales on the standard side, with the exception of the CIF Scale, shall be 25 centimeters in length. The CIF Scale shall be slightly longer.

3.1.2.2 Top runner. The top runner shall consist of a L (LOG) Scale, A (SQUARE) Scale, and a DF (FOLDED D) Scale.

3.1.2.3 Bottom runner. The bottom runner shall consist of a D Scale, DI (INVERTED D) Scale, and a K (CUBE) Scale.

3.1.2.4 Slider. The slider shall consist of a CF (FOLDED C) Scale, CIF (FOLDED INVERTED C) Scale, T (TANGENT) Scale, S (SINE-COSINE) Scale and a C Scale.

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3.1.2.5 The trigonometric scales shall be full scale, that is referenced against the C, D, and DI scales. A suitable index mark shall be placed on the C and D scales at 5.73 and marked R to eliminate the necessity of an ST scale for angles below 5.73°. Differentiation between the ascending and descending series on the S and T scales shall be accomplished by means of arrows.

*3.1.2.6 Black numbers shall appear in their correct position and with the frequency specified below:

SCALE	
L	0 thru 1 by tenths
A	1 thru 10 twice
K	1 thru 10 three times
DF, CF, CIF, C, D, DI	1 thru 2 by tenths and 2 thru 10
T	84/6, 83/7, 82/8, 81/9, 80/10 75/15, 70/20, 65/25, 60/30, 55/35, 50/40, and 45
S	84/6, 83/7, 82/8, 81/9, 80/10, 75/15, 70/20, 65/25, 60/30, 50/40, 40/50, 30/60, 20/70, and 90

3.1.3 Graduations. The width of graduation marks shall be 0.005 ± 0.001 inch. Graduation lines shall be black, straight, uniform, parallel and free from breaks or ragged edges. The graduation lines shall be accurately placed so that local or accumulated error will not affect the results of computation in excess of 0.1 of 1 percent in three significant figures.

*3.1.3.1 The graduation on the aerial side of the slide rule for each individual logarithmic scale shall be as follows:

RANGE	NUMBER OF GRADUATIONS
1-2	20
2-3	10
3-4	10
4-5	10
5-6	10
6-7	10
7-8	5
8-9	5
9-10	5
	5

The above table shall hold true in all cases except where these graduations interfere with the special mark called out in 3.1.1.2, 3.1.1.3.1, and 3.1.1.3.2.

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*3.1.3.2 The graduations on the standard side of the slide rule shall be as follows:

SCALE	NUMBER OF DIVISION MARKS
L	501
A	321
DF	321
CF	321
CIF	323
T	362
S	318
C	321
D	321
DI	321
K	271

3.2 Material. The slide rule, end plates, rivets, and screws shall be constructed of a 6061 aluminum, or a suitable substitute approved by the procuring activity. The cursor bars shall be made of nylon. The leaf spring shall be made of phosphor bronze coated with a 0.0003 inch coating of cadmium in accordance with QQ-P-416, or properly tempered 302 stainless steel.

3.3 Surface. Both faces of the slide rule shall be surfaced with baked enamel or equivalent. The faces and markings shall withstand the abrasion encountered in usage without chipping or peeling.

3.4 Color. Facings shall be either white or yellow-green (eye saver). The yellow-green shall fall between color chips 23814 and 24552 of FED-STD-595 (approximately 5800 Angstrom units).

3.5 Cursor. The cursor windows shall be tempered glass or plastic. The hairline shall be black, clearly legible, 0.003 ± .001 inches thick and in such a position that when assembled, it shall be parallel to the graduation marks on the faces. The line shall run the entire width of the slide rule and be cut in the side of the glass or plastic nearest the face of the slide rule. Construction shall be such that it must be disassembled to be removed from the slide rule.

3.6 Carrying case. A sturdy leather carrying case shall be furnished and provisions shall be made for printing a table of equivalents either directly on the carrying case, or on a card permanently attached to the carrying case and protected from destruction through normal handling by being enclosed under a stitched or laminated transparent, clear plastic window.

3.7 Alignment. To allow for alignment, one runner shall be permanently attached to each end plate by 2 rivets. The other runner shall be attached to each end plate by an adjustable device, preferably by means of a screw and internally threaded tubular, oval headed rivet, which shall pass through an oversized hole in the runner.

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3.8 Identification markings. The slide rule and the case shall be identified in accordance with MIL-STD-130.

*3.9 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

*3.10 Unless otherwise specified, the contractor shall provide instructions with each slide rule (see 6.2).

*3.11 Workmanship. The slide rule shall be clean and free from any defects that could impair its accuracy, serviceability, or appearance. The finish shall be such that the graduations can be read with a minimum of reflective interference. The slide rule shall have no rough edges, burrs, cracks, chips, or scratches.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examination tests. The examination and tests required to assure conformance of those items or lots of items to be offered for acceptance, are specified in 4.3.

4.2.1 Sampling. Sampling for the slide rules, shall be conducted in accordance with Table I and II of MIL-STD-105, under general inspection level of II, with an Acceptable Quality Level (AQL) of 2.5 for major characteristics and an AQL of 4.0 for minor characteristics.

4.3 Inspection test methods

4.3.1 Classification of defects. The slide rule shall be inspected for the following:

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR
Finish	Enamel Finish and Color not as specified	X	
	Scratches, scuffs, chips and gouges on either the face or the back of the rule	X	

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EXAMINE	DEFECTS	CLASSIFICATION MAJOR	MINOR
	Dust, dirt, grease or other foreign matter imbedded in finish		X
	Hazy or cloudy areas, streaks bubbles, hairlines, or other defects which affect the appearance and serviceability of the item		X
Markings and Graduation	Not graduated as specified	X	
	Not clear, sharp and legible	X	
Scales	Not oriented and indexed as specified	X	
	Scales omitted	X	
Scales (cont)	Scales not identified as specified	X	
	Notations missing	X	
	Notations not as specified	X	
Cursor	Windows not clearly transparent i.e., foggy, yellowed, checked, or cracked	X	
	Hairline not accurately centered		X
	Hairline not perpendicular to indicator guides and parallel to scale markings	X	
	Hairline not cut in inside face of windows	X	
	Slide not spring mounted	X	
	Cursor doesn't have free and easy movement	X	
	Hairline incapable of indexing extreme ends of scales	X	
Adjustment	Incapable of adjustment of slide friction or alignment	X	
	Not equipped with adjusting screws	X	

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EXAMINE	DEFECTS	CLASSIFICATION MAJOR	MINOR
Data Strip	Not as specified		X
Markings for Identification	Missing		X
	Defective, i.e., illegible, wrong location, incomplete or incorrect		X
Workmanship	Parts missing		X
	Not constructed as specified		X
	Material not as specified		X

4.3.2 Detailed examination. The slide rules representing a lot shall be examined with particular attention given to the following detail:

a. Accuracy of dimensions

b. Clarity of calibrations

c. Efficient operation of slide and cursor

4.3.2.1 To determine accuracy of the dimensions and calibration of the aerial photo slide rule, the following two problems are to be solved five times by use of the slide rule. The average of the five solutions may not be more than two percent in error from correct answers given herein.

4.3.2.1.1 Scale. Find the scale of photography when a camera with a 36 inch Focal Length takes a vertical aerial photograph from 1,000 feet in altitude. Solution: Place 36 on Focal Length (Inches) Scale opposite 1,000 on Altitude (Feet) Scale. Read Scale of photography as 333 opposite the SCALE arrow located on the Focal Length (Inches) Scale.

4.3.2.1.2 Interval. Find the time interval required for 60 percent overlaps on 9 x 9 photographs taken at an altitude of 26,000 Ft with a camera of 36 inch focal length when the ground speed of the aircraft is 340 miles per hour. Solution: Place 36 on Focal Length (Inches) Scale opposite 26,000 on Altitude (Feet) Scale and set hairline of the cursor directly over 12, (Scale) arrow on Focal Length Scale. Without moving cursor, move slide up until M.P.E. arrow is under the hairline. On the Ground Speed Scale, move the hairline of the cursor beside the speed of the aircraft, 340 miles per hour. Under the hairline, read the image motion of the Image Size (cm) Scale to be 0.69 inches/sec. Move the slide down until 0.69 on the Image Size (cm) Scale is opposite 60% O.L. on 9" marker on the Ground Size (Feet) Scale. To find the time interval, read 5.2 seconds on the Ground Size (Feet) Scale opposite the Cycling Time Marker of the Image Size (cm) Scale.

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4.4 Inspection and testing of preparation for delivery

4.4.1 Quality conformance inspection of pack

4.4.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment, shall be considered a unit of product.

4.4.1.2 Lot. A lot shall consist of all completed packs of any one level offered for examination at one time.

4.4.1.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.4 Examination. Samples selected in accordance with 4.4.1.3 shall be examined for the following defects. AQL shall be 2.5 percent defective.

a. Materials, methods, and containers, not as specified for Level A or B. Each incorrect material, method, or container shall constitute one defect.

b. Intermediate packaged quantities not as specified for Level A or B.

c. Gross weight exceeds 200 pounds for Level A.

d. Strapping not zinc coated for Level A.

e. Gross weight exceeds the weight limitation of the box specification for Level B.

f. Boxes not reinforced with tape as specified for Level B.

g. Marking illegible, incorrect or incomplete for Level A, B, or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each case containing the slide rule shall be preserved in accordance with MIL-P-116, Method 1C-1.

5.1.1.1 Intermediate packaging. The preserved slide rules shall be intermediate packaged in quantities of 10, in a close fitting box conforming to PPP-B-636, W6c, style optional.

5.1.2 Level B. Each case containing the slide rule shall be preserved in accordance with MIL-P-116, Method III.

5.1.2.1 Intermediate packaging. The preserved slide rules shall be intermediate packaged in quantities of 10, in a close fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, style optional.

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5.1.3 Level C. Each slide rule contained in its case shall be preserved and packaged to afford protection against deterioration and physical damage during shipment from supply source to first receiving activity for immediate use.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. The slide rules, preserved and packaged as specified in 5.1, shall be packed in close fitting boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, Class 2, style optional. The gross weight of each box shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix of the applicable box specification. Strapping shall be zinc coated.

5.2.2 Level B. The slide rules, preserved and packaged as specified in 5.1, shall be packed in close fitting boxes conforming to PPP-B-636, V3c, style optional. The gross weight shall not exceed the weight limitation of the box specification. The boxes shall be reinforced with bands or reinforced filament tape, applied in accordance with the appendix to the box specification.

5.2.3 Level C. The slide rules, preserved and packaged as specified in 5.1, shall be packed, to assure carrier acceptance and safe delivery to destination at lowest rate, in fiberboard containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or with other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

• 6.1 Intended use. The LM-221A slide rule is intended for use in interpretation of Aerial Photographs.

• 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. Stock number.

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d. Slide rule instructions required, unless otherwise specified (see 3.10).

e. Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation, packaging, and packing is intended to provide economical but limited protection, and should be specified only when it is determined the slide rules will be held in covered storage.

*6.3 The margin of this specification is marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - AS
Air Force - 70

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